Off Axis

$\vec{r}(x,y,z) \longleftrightarrow \vec{r}(s,\varphi,z)$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\int_V \frac{\vec{J}_b(\vec{r}')}{\mathscr{r}}\,d\tau' + \frac{\mu_0}{4\pi}\oint_S \frac{\vec{K}_b(\vec{r}')}{\mathscr{r}}\,d\vec{a}'$$, assuming $\vec{M}$ is uniform

$\mathscr{r} = |\vec{r}-\vec{r}'|$, $\nabla\times\vec{M} = J_b = 0$, $\vec{K}_b = \vec{M}\times\hat{n} \longrightarrow$ 3 surfaces $\longrightarrow$ did this for on-axis $\longrightarrow \vec{K}_b = M\hat{\varphi}'$

$\vec{r} = s\cos\varphi\,\hat{x} + s\sin\varphi\,\hat{y} + z\hat{z}$ $\longrightarrow$ can simplify because rotating about z-axis (angle φ) nothing changes, therefore $\vec{r}(s,0,z)$

$\vec{r}' = R\cos\varphi'\,\hat{x} + R\sin\varphi'\,\hat{y} + z'\hat{z}$

$\vec{r} = s\cos(0)\,\hat{x} + s\sin(0)\,\hat{y} + z\hat{z} = s\hat{x} + z\hat{z}$

$$|\vec{r}-\vec{r}'| = \sqrt{s^2+R^2-2sR\cos\varphi'+(z-z')^2}$$

$\vec{r}-\vec{r}' = (s-R\cos\varphi')\,\hat{x} + R\sin\varphi'\,\hat{y} + (z-z')\hat{z}$

$$|\vec{r}-\vec{r}'|^2 = (s-R\cos\varphi')^2 + R^2\sin^2\varphi' + (z-z')^2 = s^2-2sR\cos\varphi'+R^2\cos^2\varphi'+R^2\sin^2\varphi'+(z-z')^2$$
$$= s^2-2sR\cos\varphi'+R^2+(z-z')^2$$

$$\int_0^{2\pi} \frac{\sin\varphi'}{|\vec{r}-\vec{r}'|}\,d\varphi' = 0 \text{ (symmetric)}$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi}\oint \frac{M(-\sin\varphi'\,\hat{x}+\cos\varphi'\,\hat{y})}{\sqrt{s^2+R^2-2sR\cos\varphi'+(z-z')^2}}\,d\vec{a}' = \frac{\mu_0 MR}{4\pi}\int_0^{2\pi}\int_{-L/2}^{L/2} \frac{-\sin\varphi'\,\hat{x}+\cos\varphi'\,\hat{y}}{|\vec{r}-\vec{r}'|}\,dz'\,d\varphi'$$

$$\vec{A}(\vec{r}) = \frac{\mu_0 MR}{4\pi}\hat{y}\int_{-L/2}^{L/2}\int_0^{2\pi} \frac{\cos\varphi'}{|\vec{r}-\vec{r}'|}\,d\varphi'\,dz' \longrightarrow \text{tackle } dz' \text{ integral first}$$

$$\int \frac{dz'}{\sqrt{c^2+(z-z')^2}} \qquad u = z'-z,\quad du = dz' \longrightarrow \int\frac{du}{\sqrt{c^2+u^2}} \qquad u = c\tan\theta,\quad du = c\sec^2\theta\,d\theta = \int\frac{c\sec^2\theta\,d\theta}{c\sqrt{1+\tan^2\theta}} = \int\sec\theta\,d\theta = \int\frac{\sec\theta(\sec\theta+\tan\theta)}{\sec\theta+\tan\theta}\,d\theta = \ln(\sec\theta+\tan\theta)$$

derivative of top is bottom

$c^2 = s^2+R^2-2sR\cos\varphi'$

$$u = c\tan\theta \longrightarrow \tan\theta = \frac{u}{c} \longrightarrow \sec\theta = \frac{\sqrt{u^2+c^2}}{c} \longrightarrow \ln(\sec\theta+\tan\theta) = \ln\left|\frac{\sqrt{u^2+c^2}}{c}+\frac{u}{c}\right| = \ln\left(\sqrt{u^2+c^2}+u\right) - \cancel{\ln(c)}^{\,0}$$

$$\int_{u=-(L/2+z)}^{u=L/2-z}\frac{du}{\sqrt{c^2+u^2}} = \ln\left(u+\sqrt{u^2+c^2}\right)\Big|_{u=-(\frac{L}{2}+z)}^{u=\frac{L}{2}-z} = \underbrace{\ln\left(\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}-z\right)^2+c^2}\right)}_{f_1} - \underbrace{\ln\left(-\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}+z\right)^2+c^2}\right)}_{f_2} = \ln\left(\frac{\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}-z\right)^2+c^2}}{-\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}+z\right)^2+c^2}}\right)$$

$$\int_{-L/2}^{L/2}\frac{dz'}{\sqrt{s^2+R^2-2sR\cos\varphi'+(z-z')^2}} = \ln\left(\frac{\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}-z\right)^2+s^2+R-2sR\cos\varphi'}}{-\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}+z\right)^2+s^2+R^2-2sR\cos\varphi'}}\right)$$

$$\vec{A}(\vec{r}) = \frac{\mu_0 MR}{4\pi}\hat{y}\int_0^{2\pi}\ln\left(\frac{\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}-z\right)^2+s^2+R-2sR\cos\varphi'}}{-\frac{L}{2}-z+\sqrt{\left(\frac{L}{2}+z\right)^2+s^2+R^2-2sR\cos\varphi'}}\right)d\varphi'$$

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$$\vec{B} = \nabla \times \vec{A}, \quad B_s = -\frac{\partial A_\varphi}{\partial z}, \quad B_z = \frac{1}{s}\frac{\partial (sA_\varphi)}{\partial s}, \quad \frac{\partial}{\partial z}\ln\left(\frac{f_1}{f_2}\right) = \frac{\partial}{\partial z}\left(\ln(f_1) - \ln(f_2)\right)$$

$$B_s = -\frac{\partial A_\varphi}{\partial z} = -\frac{\mu_0 MR}{4\pi}\int_0^{2\pi} \cos\varphi' \frac{\partial}{\partial z}\left(\ln\left(\frac{f_1}{f_2}\right)\right) d\varphi', \quad \alpha^2 = \left(\frac{L}{2} - z\right)^2 + s^2 + R - 2sR\cos\varphi', \quad f_1 = \frac{L}{2} - z + \alpha$$

$$\frac{\partial}{\partial z}\ln(f_1) = \frac{\partial}{\partial f_1}\cdot\frac{\partial f_1}{\partial z} = \frac{1}{f_1}\frac{\partial f_1}{\partial z}, \quad \frac{\partial f_1}{\partial z} = -1 + \frac{\partial \alpha}{\partial z}, \quad \frac{\partial \alpha}{\partial z} = \frac{1}{2}(\alpha^2)^{-1/2}\left(2\left(\frac{L}{2} - z\right)(-1)\right) = \frac{-\left(\frac{L}{2} - z\right)}{\alpha} = \frac{z - \frac{L}{2}}{\alpha}$$

$$\frac{\partial f_1}{\partial z} = -1 - \frac{\left(\frac{L}{2} - z\right)}{\alpha} = \frac{-\alpha - \frac{L}{2} - z}{\alpha} = \frac{-f_1}{\alpha}, \quad \frac{\partial}{\partial z}\ln(f_1) = \frac{1}{f_1}\cdot\frac{-f_1}{\alpha} = \frac{-1}{\alpha}, \quad \frac{\partial}{\partial z}\ln(f_1) = \frac{-1}{\sqrt{\left(\frac{L}{2} - z\right)^2 + s^2 + R^2 - 2sR\cos\varphi'}}$$

$$\beta^2 = \left(\frac{L}{2} + z\right)^2 + s^2 + R - 2sR\cos\varphi', \quad f_2 = -\frac{L}{2} - z - \beta$$

$$\frac{\partial f_2}{\partial z} = -1 + \frac{\partial \beta}{\partial z}, \quad \frac{\partial \beta}{\partial z} = \frac{\frac{L}{2} + z}{\beta} \longrightarrow \frac{\partial f_2}{\partial z} = -1 + \frac{\left(\frac{L}{2} + z\right)}{\beta} = \frac{-\beta + \left(\frac{L}{2} + z\right)}{\beta} = \frac{-f_2}{\beta} \longrightarrow \frac{\partial}{\partial z}\ln(f_2) = \frac{-1}{\beta} = \frac{-1}{\sqrt{\left(\frac{L}{2} + z\right)^2 + s^2 + R^2 - 2sR\cos\varphi'}}$$

$$\frac{\partial}{\partial z}\ln\left(\frac{f_1}{f_2}\right) = \frac{-1}{\alpha} + \frac{1}{\beta}$$

$$B_s = \frac{\mu_0 MR}{4\pi}\int_0^{2\pi} \cos\varphi' \left(\frac{1}{\sqrt{\left(\frac{L}{2} - z\right)^2 + s^2 + R^2 - 2sR\cos\varphi'}} - \frac{1}{\sqrt{\left(\frac{L}{2} + z\right)^2 + s^2 + R^2 - 2sR\cos\varphi'}}\right) d\varphi' \longrightarrow$$

each term inside integral takes form

$$\int_0^{2\pi} \frac{\cos\varphi'}{\sqrt{A - B\cos\varphi'}} d\varphi', \text{ where } A = \left(\frac{L}{2} \pm z\right)^2 + s^2 + R^2, \quad B = 2sR$$

(elliptic integrals)

$$\int_0^{2\pi} \frac{\cos\varphi'}{\sqrt{A - B\cos\varphi'}} d\varphi', \quad \cos\varphi' = \frac{A - (A - B\cos\varphi')}{B} \longrightarrow \frac{\cos\varphi'}{\sqrt{A - B\cos\varphi'}} = \frac{A - (A - B\cos\varphi')}{B\sqrt{A - B\cos\varphi'}}$$

$$= \frac{A - (A - B\cos\varphi')}{B\sqrt{A - B\cos\varphi'}} = \frac{A}{B\sqrt{A - B\cos\varphi'}} - \frac{A - B\cos\varphi'}{B\sqrt{A - B\cos\varphi'}} = \frac{1}{B}\left(\frac{A}{\sqrt{A - B\cos\varphi'}} - \sqrt{A - B\cos\varphi'}\right) \longrightarrow \int_0^{2\pi} \frac{\cos\varphi'}{\sqrt{A - B\cos\varphi'}} d\varphi' = \frac{1}{B}\left[\underbrace{A\int_0^{2\pi} \frac{d\varphi'}{\sqrt{A - B\cos\varphi'}}}_{I_1} - \underbrace{\int_0^{2\pi} \sqrt{A - B\cos\varphi'}\, d\varphi'}_{I_2}\right]$$

$$I_1 = \int_0^{2\pi} \frac{d\varphi'}{\sqrt{A - B\cos\varphi'}}, \quad \cos\varphi' = 1 - 2\sin^2\left(\frac{\varphi'}{2}\right), \quad I_1 = \frac{4}{\sqrt{A+B}} K(k), \quad k^2 = \frac{2B}{A+B}, \quad I_2 = 4\sqrt{A+B}\, E(k)$$

$$I_2 = \int_0^{2\pi} \sqrt{A - B\cos\varphi'}\, d\varphi', \quad \cos\varphi' = 1 - 2\sin\left(\frac{\varphi'}{2}\right), \quad \theta = \frac{\varphi'}{2}, \quad d\varphi' = 2d\theta \longrightarrow I_2 = 2\int_0^{\pi} \sqrt{A - B + 2B\sin^2\theta}\, d\theta = 2\sqrt{A+B}\int_0^{\pi} \sqrt{1 - k^2\cos^2\theta}\, d\theta = 4\sqrt{A+B}\int_0^{\pi/2} \sqrt{1 - k^2\sin^2\psi}\, d\psi$$

$$\int_0^{2\pi} \frac{\cos\varphi'}{\sqrt{A - B\cos\varphi'}} d\varphi' = \frac{4}{B\sqrt{A+B}}\left[AK(k) - (A+B)E(k)\right], \quad k^2 = \frac{2B}{A+B} = \frac{4sR}{\left(\frac{L}{2} \pm z\right)^2 + (s+R)^2}$$

- now put B_s in terms of elliptic integrals
- do the exact same thing for B_z
- numerically compute B_s + B_z elliptic integrals

$\longrightarrow$ too much work $\longrightarrow$ numerically compute $\vec{A}(\vec{r})$, then compute $\vec{B} = \nabla \times \vec{A}$ numerically

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$$A_\varphi = \frac{\mu_0 M R}{4\pi} \int_0^{2\pi} \int_{-L/2}^{L/2} \frac{\cos\varphi'}{\sqrt{s^2+R^2-2sR\cos\varphi'+(z-z')^2}}\, dz'\, d\varphi' \longrightarrow \text{non-dimensionalize}$$

$s = \sigma L$

$R = \eta L$

$z = \zeta L$

$z' = \zeta' L, \; dz' = L\, d\zeta'$

$z' = \pm L/2 \rightarrow \zeta' = \pm 1/2$

$\vec{\beta} = \frac{\vec{B}}{\mu_0 M} \longrightarrow \boxed{\vec{\beta}_s = -\frac{\partial \bar{A}_\varphi}{\partial \zeta}, \quad \vec{\beta}_z = \frac{1}{\sigma}\frac{\partial(\sigma \bar{A}_\varphi)}{\partial \sigma}} \longrightarrow$ now this only depends on $\eta = \frac{R}{L}$ (same as on-axis problem)

$$\boxed{\bar{A}_\varphi = \frac{\eta}{4\pi} \int_0^{2\pi} \int_{-1/2}^{1/2} \frac{\cos\varphi'}{\sqrt{\sigma^2+\eta^2-2\sigma\eta\cos\varphi'+(\zeta-\zeta')^2}}\, d\zeta'\, d\varphi'} \longrightarrow$$

$$B_s = -\frac{\partial A_\varphi}{\partial z} = -\frac{\mu_0 M}{L}\frac{\partial(\mu_0 M L \bar{A}_\varphi)}{\partial \zeta} \cdot \frac{1}{L}$$

$$= -\mu_0 M \frac{\partial \bar{A}_\varphi}{\partial \zeta}$$

$$B_z = \frac{1}{s}\frac{\partial(sA_\varphi)}{\partial s} = \frac{\mu_0 M}{\sigma}\frac{\partial(\sigma \bar{A}_\varphi)}{\partial \sigma}$$

- wrote a python program
- uses scipy's QUADPACK numerical method
- fills magnet area with a uniform field
- applys off-axis numerical solution to axis points in space
- applys on-axis analytical expression when $\sigma = 0$

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